

Day 89: Basic Algebra for Machine Learning

1. Day Number

Day 89

2. Topic Name

Variables, Equations, Functions, and Simple Linear Relationships

Today you will learn some school-level algebra that appears frequently in machine learning.

The main ideas are:

- variable
- constant
- equation
- coefficient
- function
- linear equation

No advanced mathematics is needed.

3. Connection

You already use **variables in Python**:

```
distance = 5  
price = 100
```

In mathematics, variables have a similar purpose.

For example:

$$x = 5$$

Here, x represents a value.

Machine learning formulas often use variables such as:

```
x → input  
y → output
```

For example:

```
x = size of a house  
y = estimated house price
```

Today you will learn how an input can be mathematically connected to an output.

4. Important Topics

Variable

A **variable** represents a value that can change.

For example:

$$x = \text{distance travelled}$$

If different customers live at different distances, x might be:

1 km
3 km
5 km
10 km

In Python, the idea is similar:

```
x = 5
```

Constant

A **constant** is a fixed value.

Suppose every delivery has a fixed starting charge of ₹40.

Then:

```
40 = constant
```

It stays the same even when the distance changes.

Equation

An **equation** says that two mathematical expressions are equal.

For example:

$$y = 10x + 40$$

It describes how y depends on x .

Coefficient

A **coefficient** is a number multiplying a variable.

For example:

$$y = 10x + 40$$

The coefficient of x is:

10

because:

$$10x = 10 \times x$$

Function

A **function** takes an input and produces an output.

For example:

```
input distance
  ↓
delivery-cost function
  ↓
output cost
```

You might write:

$$f(x) = 10x + 40$$

If x changes, the function calculates a new result.

This is similar to a Python function:

```
def delivery_cost(x):
    return 10 * x + 40
```

Linear Equation

A simple linear equation has a relationship that changes at a constant rate.

One common form is:

$$y = mx + b$$

where:

```
x = input
y = output
m = coefficient / slope
b = constant / intercept
```

5. Foundational Notes

When learning machine learning mathematics, do not think of the letters as mysterious symbols.

Ask what each letter represents.

For example:

x = delivery distance
y = delivery cost
m = cost added for every kilometre
b = starting delivery charge

Then the formula becomes much easier to understand.

A useful translation is:

Mathematics:
 $y = mx + b$

Plain English:
output =
(rate × input)
+ starting amount

6. Understanding $y = mx + b$

Suppose a delivery company charges:

Base delivery charge = ₹40
Additional charge = ₹10 per kilometre

Let:

x = distance in kilometres
y = total delivery cost

Here:

m = 10
b = 40

The relationship is:

In this example:

x

The input.

x = distance

y

The output we want to calculate.

y = delivery cost

m

The amount y changes when x increases by one.

m = ₹10 per kilometre

b

The starting value when $x = 0$.

b = ₹40

So even for zero kilometres, the model starts with the ₹40 base charge.

7. Easy Numerical Example

Suppose:

$x = 5$ km
 $m = 10$
 $b = 40$

Substitute the numbers:

$$y = 10(5) + 40$$

First multiply:

$$10 \times 5 = 50$$

Then add the constant:

$$50 + 40 = 90$$

Therefore:

For 5 km:

Delivery cost = ₹90

In Python, the same calculation could be written as:

```
x = 5
m = 10
b = 40

y = m * x + b
```

The mathematical and Python ideas are almost identical.

8. Problem Statement

A delivery company uses the following rule:

Base delivery charge = ₹30
Cost per kilometre = ₹8

Let:

x = delivery distance
 y = total delivery cost

Use the linear relationship to calculate the delivery cost for:

$x = 2$ km
 $x = 5$ km
 $x = 10$ km

Your goal is to identify:

$m = ?$
 $b = ?$

and then substitute each value of x .

9. Concepts Used

This exercise uses:

- variable
- input
- output
- constant
- coefficient
- multiplication
- addition
- equation
- function
- linear relationship

The important interpretation is:

$x \rightarrow$ distance
 $m \rightarrow$ price per kilometre
 $b \rightarrow$ base charge
 $y \rightarrow$ final delivery cost

10. Thought Process

When you see a formula such as a linear relationship, do not immediately calculate.

First identify what everything means.

Step 1: Identify the input

Here:

$x =$ distance

Step 2: Identify the coefficient

Ask:

| How much does the cost increase for each kilometre?

That number becomes m .

Step 3: Identify the constant

Ask:

| What amount must be paid regardless of distance?

That becomes b .

Step 4: Substitute x

For each distance, replace x with the given number.

Step 5: Multiply first

Calculate:

$m \times x$

Step 6: Add the constant

Finally:

$(m \times x) + b$

That gives y .

11. Beginner-Friendly Mathematical Steps

For every input value, follow the same pattern:

1. Write the formula.
2. Identify m .
3. Identify b .
4. Replace x with the input value.
5. Multiply $m \times x$.
6. Add b .
7. The answer is y .

For example, with different practice numbers:

```
m = 4
b = 20
x = 3
```

Start with:

$$y = 4(3) + 20$$

Multiply:

$$4 \times 3 = 12$$

Add:

$$12 + 20 = 32$$

Therefore:

```
x = 3
y = 32
```

12. Connection to Future Linear Regression

This algebra becomes very important when you learn **Linear Regression**.

Imagine a dataset containing:

House Size	House Price
800	₹...

1000	₹...
1200	₹...
1500	₹...

You might use:

```
x = house size
y = predicted house price
```

A simple linear regression model tries to learn a relationship similar to:

```
predicted price
=
(coefficient × house size)
+
starting value
```

The important difference is that today **you are given** the coefficient and constant.

Later, in Linear Regression, the machine-learning algorithm will try to **learn suitable values from data**.

For now, simply understand how an input travels through a linear formula to produce an output.

13. Easy Edge Cases

Case 1: $x = 0$

Suppose:

```
m = 10
b = 40
x = 0
```

Then:

$$y = 10(0) + 40$$

Since:

$$10 \times 0 = 0$$

the output is simply:

```
y = 40
```

This helps explain the meaning of b .

When the input is zero, b is the starting output.

Case 2: Coefficient = 0

Suppose:

$$\begin{aligned}m &= 0 \\b &= 50\end{aligned}$$

Now changing x does not affect the result because:

$$0 \times x = 0$$

So:

$$\begin{aligned}x = 1 &\rightarrow y = 50 \\x = 5 &\rightarrow y = 50 \\x = 20 &\rightarrow y = 50\end{aligned}$$

The output remains constant.

14. Expected Calculations

For your delivery-cost exercise, your working should have this structure:

Distance (x)	Calculation	Delivery cost (y)
2 km	$(\text{cost per km} \times 2) + \text{base charge}$	Calculate
5 km	$(\text{cost per km} \times 5) + \text{base charge}$	Calculate
10 km	$(\text{cost per km} \times 10) + \text{base charge}$	Calculate

You should notice that as distance increases, the delivery cost also increases at a **constant rate**.

For every additional kilometre, the output increases by the same amount.

That constant rate is the **coefficient**.

15. Hint Only

For the exercise:

$$\begin{aligned}\text{Base charge} &= ₹30 \\ \text{Cost per km} &= ₹8\end{aligned}$$

Ask yourself:

Which number is multiplied by x ?
→ that is m

Which number stays fixed?
→ that is b

Then, for $x = 2$, begin by substituting the values into the linear formula.

Repeat exactly the same process for:

$x = 5$
 $x = 10$

Try to calculate the three final values yourself.

The key lesson for **Day 89** is:

A linear relationship connects an input and an output using a constant rate of change and a starting value. Understanding this simple algebra prepares you for Linear Regression, where a model learns those values from data.